

## PH 550: Introduction to Quantum Optics

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| Course Code      | PH 550                                                                                     |
| Course Name      | Introduction to Quantum Optics                                                             |
| L-T-P-C          | 3-0-0-3                                                                                    |
| Prerequisites    | Basics of Quantum Mechanics , Classical Electrodynamics ,Mathematical Methods for Physics  |
| Intended for     | UG/PG/IPhD/PhD                                                                             |
| Mutual Exclusion | None                                                                                       |
| Approval         | 57th BoA (Oct 2024); 47th Senate (Sep 2025), in the Minor in Quantum Technology curriculum |

### Course Contents

- **Module 1: Two-Level Atom System** (8 lectures)
  - Oscillating dipoles and Bloch sphere representation
  - Density Operator and Density Matrix formalism
  - Optical Bloch Equations
  - Rabi Frequency and Spontaneous Decay
  - Lineshape in Fluorescence and Saturation Broadening
- **Module 2: Quantization of the Electromagnetic Field** (14 lectures)
  - Quantized Modes and Operators
  - Ladder Operators, Number Operator, and Quadrature Operators
  - Quantization of the Electromagnetic Field
  - Hamiltonian of Radiation Field and Vacuum State Energy
  - Fock States, Coherent States, and Squeezed States
  - Thermal States and Planck's Black Body Radiation Formula
- **Module 3: Photon Statistics and Measurement** (10 lectures)
  - Photon Statistics
    - \* Poissonian, Sub-Poissonian, and Super-Poissonian light
    - \* Intensity interferometer and Hanbury Brown–Twiss experiments
    - \* Second-order correlation function,  $g_2(t)$
    - \* Photon bunching and antibunching phenomena
    - \* Coherent, Bunched, and Antibunched light
  - Experimental Demonstrations
    - \* Photon antibunching experiments
    - \* Single-photon sources and their application in quantum technologies
- **Module 4: Atoms in Cavities** (10 lectures)
  - Optical and Microwave Cavities
    - \* Fundamentals of Optical Cavities
    - \* Microwave Cavities and Rydberg Atoms
  - Atom-Cavity Coupling
    - \* Weak coupling regime and the Purcell Effect
    - \* Experimental observations of the Purcell Effect
    - \* Strong coupling regime and Cavity QED
    - \* Experimental observation of strong coupling in cavities

### Laboratory Modules

1. NA

### Text Books

1. Scully, M. O., & Zubairy, M. S., **Quantum optics**.
2. Christopher Gerry and Peter Knight, **Introductory Quantum Optics**.
3. Mark Fox, **Quantum Optics**.
4. Leonard Mandel and Emil Wolf, **Optical Coherence and Quantum Optics**.

### References

1. NA